

Lecture: 16 Asymptotes

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Definition: Asymptote

A straight line at a finite distance from the origin to which a tangent to a curve tends, as the distance from the origin of the point of contact tends to infinity, is called an asymptote of the curve. The tangent to the curve $y = f(x)$ at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x) \Rightarrow Y = \frac{dy}{dx}X + \left(y - x\frac{dy}{dx}\right).$$

Excluding for the present asymptotes parallel to the y -axis, above equation shows that as $x \rightarrow \infty$

$$\frac{dy}{dx} \text{ and } y - x\frac{dy}{dx},$$

must both tend to a finite limits, say m and c , in order that an asymptote might exist. If this condition is satisfied, the asymptote would be

$$y = mx + c.$$

As we get that $y = f(x)$ has an asymptote which is not parallel to the y -axis, then

$$\lim_{x \rightarrow \infty} \left(y - x\frac{dy}{dx}\right) \text{ is finite.}$$

It follows that

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{y - x\frac{dy}{dx}}{x} &= 0 \\ \Rightarrow \lim_{x \rightarrow \infty} \frac{y}{x} - \lim_{x \rightarrow \infty} \frac{dy}{dx} &= 0 \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow \infty} \frac{y}{x} = \lim_{x \rightarrow \infty} \frac{dy}{dx} = m.$$

Rule for finding the asymptotes of some general algebraic curve.

Let the equation of curve be

$$a_0y^n + a_1y^{n-1}x + a_2y^{n-2}x^2 + \dots + a_{n-1}yx^{n-1} + a_nx^n \quad (1)$$

$$+ b_1y^{n-1} + b_2y^{n-2}x + \dots + b_nyx^{n-2} + b_nx^{n-1} \quad (2)$$

$$+ c_2y^{n-2} + \dots + \dots = 0. \quad (3)$$

or

$$x^n\phi_n\left(\frac{y}{x}\right) + x^{n-1}\phi_{n-1}\left(\frac{y}{x}\right) + \dots = 0.$$

If we substitute $y = mx + c$ and expand, we get

$$x^n \left\{ \phi_n(m) + \frac{c}{x} \phi'_n(m) + \frac{c^2}{2!x^2} \phi''_n(m) + \dots \right\} + x^{n-1} \left\{ \phi_{n-1}(m) + \frac{c}{x} \phi'_{n-1}(m) + \dots \right\} + \dots = 0.$$

If we now equate to zero the coefficient of the two highest powers of x , we can determine m and c from these equations. If $m_1, c_1; m_2, c_2; \dots$ are the values of m and c thus obtained, the asymptotes are

$$y = m_1x + c_1; y = m_2x + c_2; \dots$$

Example. Find the asymptote of the following curve

$$x^3 + 2x^2y - xy^2 - 2y^3 + xy - y^2 = 1.$$

Solution. Putting $y = mx + c$, we have

$$\begin{aligned} & x^3 + 2x^2(mx + c) - x(mx + c)^2 - 2(mx + c)^3 + x(mx + c) - (mx + c)^2 - 1 = 0, \\ \Rightarrow & x^3(-2m^3 - m^2 + 2m + 1) + x^2(-m^2 - 6cm^2 - 2mc + m + 2c) + x(-c^2 - 6mc^2 + c - 2mc) + (-2c^3 - c^2 - 1) = 0. \end{aligned}$$

Therefore, m and c are given by

$$-2m^3 - m^2 + 2m + 1 = 0,$$

$$\text{and } c(2 - 2m - 6m^2) - (m^2 - m) = 0.$$

The first equation gives

$$(m - 1)(m + 1)(m + 1/2) = 0 \Rightarrow m = 1, m = -1, m = -1/2.$$

The second equation gives

$$c = \frac{m^2 - m}{2 - 2m - 6m^2}.$$

Therefore, when $m = 1, c = 0$; when $m = -1, c = -1$; when $m = -1/2, c = 1/2$.

Hence the asymptotes are

$$y = x; y = -x - 1; y = -\frac{1}{2}x + \frac{1}{2}.$$

Two Parallel Asymptotes

The rule for finding asymptotes can be extended as follows: Substitute $mx + c$ for y in the equation of the curve and equate to zero the coefficient of highest power of x . If any value of m derived from this equation makes the coefficient of x^{n-1} identically zero, then the corresponding value of c should be determined from the equation obtained by putting the coefficient of x^{n-2} equal to zero.

Example. Find the asymptotes of the curve

$$x^3 + x^2y - xy^2 - y^3 - 3x - y - 1 = 0.$$

Solution. Putting $y = mx + c$, we have

$$x^3 + x^2(mx + c) - x(mx + c)^2 - (mx + c)^3 - 3x - (mx + c) - 1 = 0,$$

$$x^3(1 + m - m^2 - m^3) + x^2(c - 2mc - 3mc^2) + x(-c^2 - 3mc^2 - m - 3) + (-c^3 - c - 1) = 0.$$

Therefore, m and c must satisfy

$$1 + m - m^2 - m^3 = 0,$$

$$\text{and } c - 2mc - 3mc^2 = 0.$$

The first equation gives $m = 1, -1, -1$. The second equation then gives $c = 0$, when $m = 1$. When $m = -1$, however the second equation reduces to the identity $0 \cdot c = 0$

Equating, therefore the coefficient of the next lower power of x i.e. x^2 we have

$$-c^2 - 3mc^2 - m - 3 = 0 \text{ which gives } c = \pm 1 \text{ after putting } m = -1.$$

Hence, the asymptotes are

$$y = x; y = -x + 1; y = -x - 1.$$

Asymptotes Parallel to the axes

The asymptotes parallel to x -axis can be obtained by equating to zero the coefficient of the highest power of x , provided this is not a constant.

Similarly, the asymptotes parallel to the y -axis are obtained by equating to zero the coefficient of the highest power of y , provided this is not a constant.

Example. Find the equation of asymptotes of the following curve which are parallel to either of the axis

$$x^2y^3 + x^3y^2 = x^3 + y^3.$$

Solution. The given equation can be written as

$$x^2y^3 + x^3y^2 - x^3 - y^3 = 0,$$

$$\Rightarrow (x^2 - 1)y^3 + (y^2 - 1)x^3 = 0.$$

Equating to zero the highest power of x and y , we get

$$x^2 - 1 = 0 \text{ and } y^2 - 1 = 0.$$

Hence, the required asymptotes are

$$x = \pm 1 \text{ and } y = \pm 1.$$

Shorter method for finding asymptote

$\phi_n(m)$ can be obtained at once by putting $x = 1$ and $y = m$ in the highest degree terms of the equation to the curve. Similarly, $\phi_{n-1}(m)$ can be obtained by putting $x = 1$ and $y = m$ in the $(n - 1)$ th degree terms. Hence, we get the asymptote more quickly as following:

In the highest degree terms put $x = 1$ and $y = m$, thus getting $\phi_n(m)$. Equate this to zero and solve for m . Let the roots be $m_1, m_2, \dots, m_n$. Next from $\phi_{n-1}(m)$ in a similar way from the terms of degree $n - 1$. Then, the values of $c_1, c_2, \dots, c_n$ are found by substituting $m_1, m_2, \dots, m_n$ in term in the formula

$$c = -\frac{\phi_{n-1}(m)}{\phi'_n(m)}.$$

The asymptotes then are $y = m_1x + c_1, y = m_2x + c_2, \dots, y = m_nx + c_n$.

Example. Find the asymptote of the following curve

$$4x^3 - x^2y - 4xy^2 + y^3 + 3x^2 + 2xy - y^2 = 7.$$

Solution. Putting $x = 1$ and $y = m$ in the 3rd degree terms and equating to zero, we get

$$m^3 - 4m^2 - m + 4 = 0.$$

$$\Rightarrow (m - 4)(m - 1)(m + 1) = 0 \Rightarrow m = 4, 1, -1.$$

$$\text{Then, } c = \frac{\phi_{n-1}(m)}{\phi'_n(m)} = \frac{3 + 2m - m^2}{3m^2 - 8m - 1} = \frac{1}{3}, \frac{2}{3}, 0, \text{ respectively.}$$

Therefore, the asymptotes are $y = 4x + \frac{1}{3}, y = x + \frac{2}{3}, y = -x$.

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.